Student Study & Practice Guide

Module 3: Mathematics for Artificial Intelligence

> **Module Objective:** Understand the intuitive foundation behind statistical analysis, linear algebra, vector representations, calculus derivatives, and gradient descent optimization used in modern Artificial Intelligence.

1. Basics of Statistics

Statistics is the science of collecting, organizing, analyzing, and interpreting data. Raw numbers are useless to an AI model until structured through statistical processing.

- **Descriptive Statistics:** Summarizes characteristics of a dataset (e.g., mean income, age range).

- **Inferential Statistics:** Uses sample data to draw conclusions or predictions about a larger population (e.g., election polling).

2. Measures of Central Tendency

Central values help AI models establish baseline expectations in data distributions.

```
Mean = Sum of Values / Total Number of Values = Σx / N
```

- **Mean:** Arithmetic average. Highly sensitive to extreme values (outliers).
- **Median:** Middle number in an ordered sequence. Highly resilient to skewed data.
- **Mode:** The value that appears most frequently. Ideal for categorical selections.

> **Worked Example 1: Calculating Central Tendency**
>
> **Dataset:** Marks obtained by 5 students: 10, 20, 20, 30, 40
>
> **Mean Solution:** (10 + 20 + 20 + 30 + 40) / 5 = 120 / 5 = **24**
>
> **Median Solution:** Ordered dataset: 10, 20, **20**, 30, 40 → Middle element = **20**
>
> **Mode Solution:** Value occurring most frequently = **20**

3. Standard Deviation (Data Dispersion)

Measures how widely data points deviate from the distribution mean.

```
Population SD (σ) = √[ Σ(x - μ)² / N ]
Sample SD (s) = √[ Σ(x - x⁻)² / (n - 1) ]
```

> **Real-World AI Application: Fraud Detection**

> If a bank customer typically spends ₹500 to ₹1,000 daily (Low SD), and suddenly initiates a transaction of ₹2,00,000, the extreme standard deviation spike flags an immediate fraud alert in the AI monitoring system.

4. Probability

Probability quantifies the uncertainty of events on a scale from 0 to 1 (0% to 100%).

```
P(E) = Favourable Outcomes / Total Outcomes
```

> **AI Email Spam Example**
>
> When an incoming email arrives, the spam classifier calculates: P(Spam) = 0.98 vs. P(Inbox) = 0.02. Because 0.98 > 0.50, the email is routed directly to the Spam folder.

5. Linear Algebra: Vectors & Matrices

Linear algebra provides the mathematical structure required to store, transform, and evaluate data inside computer memory and neural network layers.

- **Vector:** A 1D array of numbers. E.g., [2, 4, 6]
- **Matrix:** A 2D grid of numbers with R rows and C columns (size R × C).

> **Worked Example 2: Matrix-Vector Multiplication**
>
> Multiply Matrix A (2x2) by Vector V (2x1):
>
> A = [[1, 2], [3, 4]], V = [[2], [1]]
>
> Row 1: (1×2) + (2×1) = 2 + 2 = **4**
>
> Row 2: (3×2) + (4×1) = 6 + 4 = **10**
>
> Result Vector = **[[4], [10]]**

6. Calculus & Derivatives

Calculus studies how quantities change over continuous domains.

```
Derivative Power Rule: d/dx (x^n) = n * x^(n-1)
Example: d/dx (x²) = 2x | d/dx (3x²) = 6x | d/dx (5x³) = 15x²
```

7. Gradient Descent Optimization

Gradient descent adjusts model parameters step-by-step in the direction that minimizes loss error.

```
Parameter Update Rule: θ_new = θ_old - (α * ∂J/∂θ)
Where: θ = parameter weight, α = learning rate, ∂J/∂θ = gradient slope
```

Worked Example 3: Single Parameter Update Step

Given: Current Weight (θ) = 5, Learning Rate (α) = 0.1, Gradient Slope = 8

θ_new = 5 - (0.1 × 8) = 5 - 0.8 = **4.2**

Self-Practice Problems (Test Your Understanding)

1. **Central Tendency:** Find Mean, Median, and Mode for the dataset: 5, 7, 8, 8, 9, 12.
   (Answer: Mean = 8.17, Median = 8, Mode = 8)
2. **Standard Deviation:** Calculate SD for data points: 1, 3, 5, 7, 9.
   (Answer: Mean = 5, Variance = 8, SD = 2.83)
3. **Probability:** A bag contains 5 red, 3 blue, and 2 green balls. What is the probability of randomly drawing a blue ball?

(Answer: 3 / 10 = 0.30 or 30%)

4. **Gradient Descent:** Given weight $\theta = 10$, learning rate $\alpha = 0.05$, and gradient slope = 6. Calculate the updated weight.
   (Answer: 10 - (0.05 × 6) = 9.7)